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AI Use Statement: ChatGPT was used to assist in the development of this assignment text. ChatGPT was prompted to provide initial text for each section. A human read, reviewed, improved (at least I think the human improved), and edited the base project description to improve accuracy, transparency, and clarity as well as relevance for the course.

1 Introduction

In this assignment, you will explore finite volume methods (FVM) for solving two-dimensional convection (and if you choose to challenge yourself, convection-diffusion) problems.

The governing convection-diffusion equation in conservative form for a scalar pollution concentration field $\phi(\mathbf{x}, t)$ is given by:

$$\frac{\partial \phi}{\partial t} + \nabla \cdot (\mathbf{u}\phi) = \nabla \cdot (D\nabla \phi), \quad (1)$$

where:

- $\phi(\mathbf{x}, t)$ represents the concentration of the quantity being conserved – in this assignment this is concentration of a pollutant,
- $\mathbf{u} = (u, v)$ is the velocity field that transports the pollutant,
- D is the diffusion coefficient (assumed to be constant),
- $\nabla \cdot (\mathbf{u}\phi)$ represents the convective transport of the pollutant,
- $\nabla \cdot (D\nabla \phi)$ represents the diffusive transport.

You can read more about the convection-diffusion equation here:

https://en.wikipedia.org/wiki/Convection-diffusion_equation

1.1 Overall Project Description: Application of the Method to a “Toy” Problem

The application problem for this assignment involves modeling the transport and spread of a chemical pollutant in groundwater. The pollutant is advected by a “background” groundwater flow field. We will imagine for this assignment, that the groundwater flow will transport the pollutant towards areas where residents and companies may have water wells. Additionally, due to diffusion, the pollutant concentration spreads over time.

A key question in this study is whether it is possible to mitigate the pollutant concentration by strategically pumping water from the domain. By implementing a numerical solver and analyzing the results, you will investigate how effective a pumping strategy could be in reducing pollution levels in this area.

1.2 Overall Project Description: Numerical Methods Tasks

Required Challenge: You will begin by constructing a finite-volume solver for the convection of a pollutant in a 2D domain. You will implement two different flux functions and compare the results. Once you have a working code, you will apply it to a simplified problem that examines the use of pumps (sinks) to remove the pollutant from the flow-field.

Optional challenge for bonus points: you will determine how to introduce/add the diffusive flux term into your numerical scheme and repeat the analysis and observe changes.

2 Domain and Initial Conditions

For this problem, you will consider a two-dimensional domain of size 6 units in the x -direction and 1 unit in the y -direction. That is,

$$0 \leq x \leq 6, \quad 0 \leq y \leq 1.$$

To ensure the conservation of the solution within the domain, you will start by imposing *periodic boundary conditions* on all four of the domain boundaries. This means that any pollutant leaving one boundary will re-enter at the opposite boundary. Mathematically, this is expressed as:

$$\phi(0, y, t) = \phi(6, y, t), \quad \forall y \in [0, 1], \forall t \geq 0, \quad (2)$$

$$\phi(x, 0, t) = \phi(x, 1, t), \quad \forall x \in [0, 6], \forall t \geq 0. \quad (3)$$

2.1 Initial Condition

At the initial time $t = 0$, the pollutant concentration $\phi(x, y, 0)$ is given by a two-dimensional Gaussian function centered at $(x_c, y_c) = (0.5, 0.5)$, with a spread (standard deviation) of $\sigma = 0.1$:

$$\phi(x, y, 0) = 0.25 * \exp \left(-\frac{(x - 0.5)^2 + (y - 0.5)^2}{2(0.1)^2} \right). \quad (4)$$

This function represents a localized initial concentration of the pollutant, which will be transported (and diffused) throughout the domain over time due to convection (and diffusion effects).

3 Project Tasks and Deliverables

Please follow the questions/steps systematically, ensuring that you provide clear explanations and justifications for your results in your submitted project work. Thank You!

1. **Question 1 (10 pts): Deriving the Finite Volume Formulation** Starting from the governing convection-diffusion equation:

$$\frac{\partial \phi}{\partial t} + \underbrace{\nabla \cdot (\mathbf{u}\phi)}_{\text{Convection}} = \underbrace{\nabla \cdot (D\nabla \phi)}_{\text{Diffusion}}$$

write step-by-step (with meaningful explanations) how the finite volume formulation below is derived for a single cell of the computational domain:

$$\frac{d\phi_{i,j}}{dt} = -\frac{1}{\Delta x} (F_{i+1/2,j}^x - F_{i-1/2,j}^x) - \frac{1}{\Delta y} (F_{i,j+1/2}^y - F_{i,j-1/2}^y) \quad (5)$$

$$+ \frac{1}{\Delta x} (G_{i+1/2,j}^x - G_{i-1/2,j}^x) + \frac{1}{\Delta y} (G_{i,j+1/2}^y - G_{i,j-1/2}^y), \quad (6)$$

Note: This could also be written with compass rose nomenclature as:

$$\frac{d\phi_{i,j}}{dt} = -\frac{1}{\Delta x} (F_e^x - F_w^x) - \frac{1}{\Delta y} (F_n^y - F_s^y) \quad (7)$$

$$+ \frac{1}{\Delta x} (G_e^x - G_w^x) + \frac{1}{\Delta y} (G_n^y - G_s^y), \quad (8)$$

where:

- F^x, F^y are the numerical **convective fluxes**,
- G^x, G^y are the numerical **diffusive fluxes**.

2. **Question 2: Computational Grid and Discretization (10 pts):** Describe how you will discretize your computational domain. Clearly indicate where the solution ϕ is evaluated/stored. Show the placement of fluxes on cell faces and describe the indexing scheme you will use for numbering cells and faces both globally (ie. what overall cell is this) and with respect to x - and y -direction indexing (similar to the NNodes matrix we used in Project 1). **You are highly encouraged to provide diagram(s) to support your description.**
3. **Question 3: Upwind Convective Flux Formulation (10 pts)** Initially, you will consider a simple(r) **Upwind Flux**. This approach introduces numerical diffusion but ensures stability. The first-order upwind flux (convective) can be calculated on each cell face as: *Note: To maintain consistency with the Video 19b homework code-along and implementation slides, I changed the equation indices below as shown in red – this is so that the west and south cell fluxes are shown, I did this to hopefully avoid some confusion.*

$$F_{i-1/2,j}^x = F_{west,j}^x = \begin{cases} u_{i-1/2,j} \phi_{i-1,j}, & u_{i-1/2,j} > 0, \\ u_{i-1/2,j} \phi_{i,j}, & u_{i-1/2,j} \leq 0. \end{cases} \quad (9)$$

$$F_{i,j-1/2}^y = F_{i,south}^y = \begin{cases} v_{i,j-1/2} \phi_{i,j-1}, & v_{i,j-1/2} > 0, \\ v_{i,j-1/2} \phi_{i,j}, & v_{i,j-1/2} \leq 0. \end{cases} \quad (10)$$

To avoid using *if statements in the code*, this could be written more generally as:

$$F_{west,j}^x = \frac{1}{2} u_{i-1/2,j} \cdot (\phi_{i,j} + \phi_{i-1,j}) - \frac{1}{2} |u_{i-1/2,j}| \cdot (\phi_{i,j} - \phi_{i-1,j}) \quad (11)$$

and:

$$F_{i,south}^y = \frac{1}{2} v_{i,j-1/2} \cdot (\phi_{i,j} + \phi_{i,j-1}) - \frac{1}{2} |v_{i,j-1/2}| \cdot (\phi_{i,j} - \phi_{i,j-1}) \quad (12)$$

Question Statement: Consider the above equations representing the upwind flux. Show that equation 9 & 11 (and thus also 10 & 12) are equivalent. Also, show how you would calculate these fluxes for a single cell.

4. **Question 4: Boundary Conditions (10 pts):** In this homework, you will examine both periodic boundary conditions as well as inflow/outflow boundary conditions. Describe how you will implement each of these boundary conditions in your code.
5. **Question 5: Pseudo Code (15 pts):** Now that you have defined the equations for a finite volume method, develop a psuedo code that will guide the development of the MATLAB code in the next section(s).
6. **Question 6: Implement an upwind flux (convection only) MATLAB Finite Volume Code (15 pts):** Implement a FVM method using the the upwind flux scheme for the convection term. Set the convection velocity field as $V_x = 2$ units/second everywhere. Run your solver using the upwind scheme and visualize the pollutant concentration over time. See Figures 1 and 2 for coarse and fine meshes with the initial conditions shown for each.

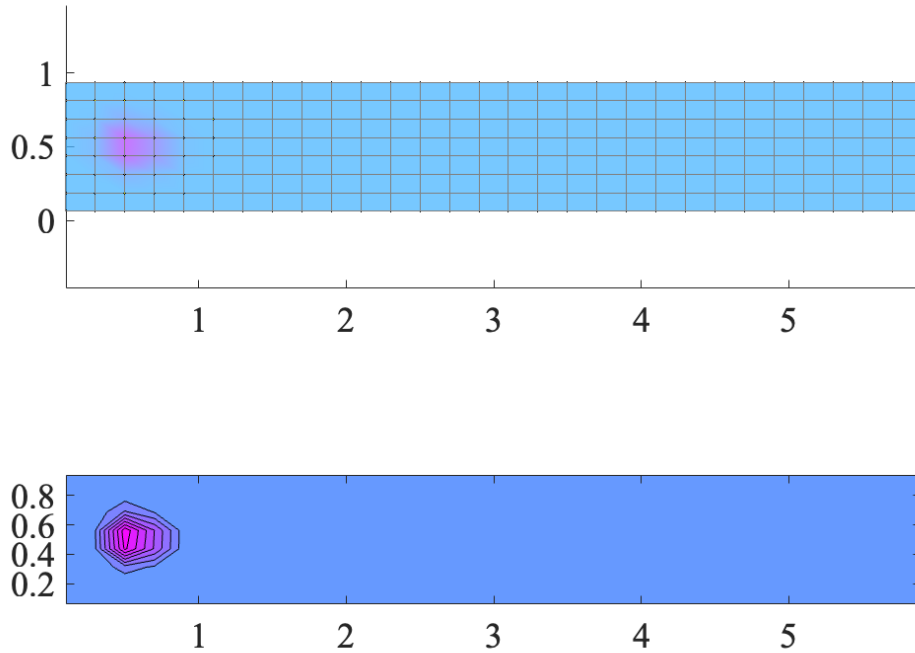


Figure 1: A course FVM mesh showing the initial condition pollution blob.

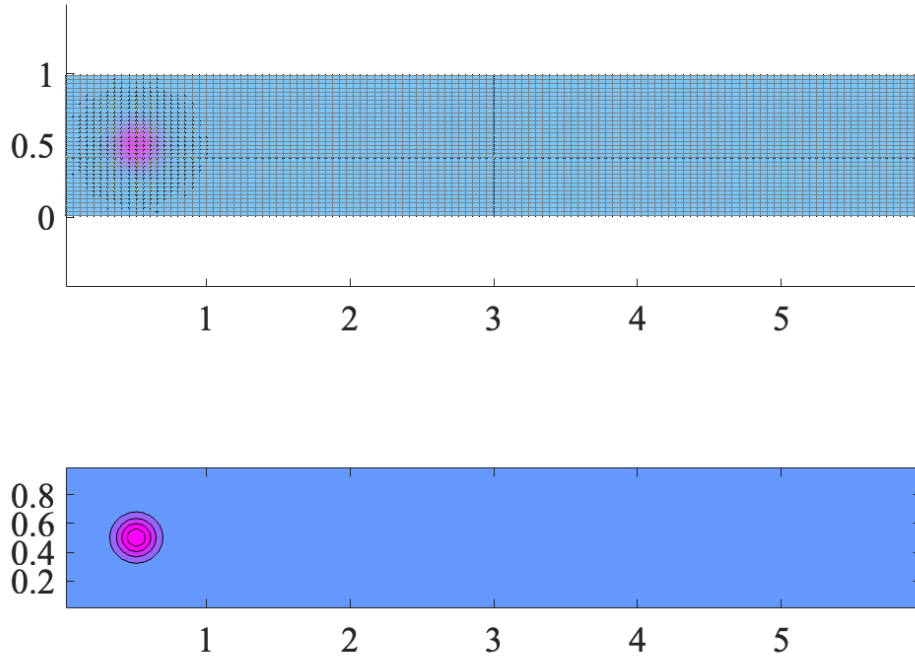


Figure 2: A medium FVM mesh showing the initial condition pollution blob.

7. **Question 6: Implement a 2D Lax-Wendroff flux (10 pts).** You’ve hopefully seen that the upwind flux was stable when the $CFL < 1$ condition was met, but, suffered from excessive diffusion for most of the grids/timesteps you examined. In this question, you will implement a 1-D Lax-Wendroff flux to try to reduce the numerical diffusion. We’ll also introduce *dimensional splitting* to allow us to easily implement this 1D flux in 2D (see bonus question #2 for the flux differencing alternative).

- (a) Modify your code to introduce an if-else condition that allows you to switch between the upwind flux and the new Lax-Wendroff flux.
- (b) We’re going to implement dimensional splitting (Chapter 19.5 of Leveque) – an easy way to extend 1D FVM approaches to multiple dimensions. The idea is to update the problem in the x -direction first, then afterwards to update the equation in the y -direction. To do this, setup your code so that you first update the pollutant concentration, ϕ using the x -direction equation for all cells:

$$\phi_{i,j}^* = \phi_{i,j}^k - \frac{\Delta t}{\Delta x} \left(F_{i+1/2,j}^k - F_{i-1/2,j}^k \right) \quad (13)$$

then, perform the second (and final) pollutant concentration update in time using the y -update equation:

$$\phi_{i,j}^{k+1} = \phi_{i,j}^* - \frac{\Delta t}{\Delta x} \left(F_{i,j+1/2}^* - F_{i,j-1/2}^* \right) \quad (14)$$

This approach approximates the effect of the diagonal propagation/cross-derivatives without the full complexity associated with a full 2-Dimensional flux function.

- (c) Try some numerical experiments to get a feeling for the behavior of the Lax-Wendroff flux scheme. What do you observe about accuracy and stability?

Lax-Wendroff Flux

The new flux scheme provides a second-order accurate flux using a Taylor series expansion in time (Leveque, Finite Volume Methods for Hyperbolic Problems, Chapter 19, equation 19.14):

$$F_{i-1/2,j}^x = u_{i-1/2,j} \left(\frac{\phi_{i-1,j} + \phi_{i,j}}{2} \right) - u_{i-1/2,j}^2 \frac{\Delta t}{2\Delta x} (\phi_{i,j} - \phi_{i-1,j}),$$

$$F_{i,j-1/2}^y = v_{i,j-1/2} \left(\frac{\phi_{i,j-1} + \phi_{i,j}}{2} \right) - v_{i,j-1/2}^2 \frac{\Delta t}{2\Delta y} (\phi_{i,j} - \phi_{i,j-1}).$$

8. **Comparison of the Convective Flux Schemes (10 pts)** In this section you will compare the two flux schemes (upwind vs. Lax-Wendroff) you have implemented in your 2D MATLAB FVM code. To do this, run the simulation until $t_f = 12$ seconds. The blob of pollutant should have convected through your domain twice.

You should run your code to explore and present some of your findings about the following:

- (a) The maximum allowable timestep to maintain a stable solution.
- (b) The grid refinement and its impact on both schemes.
- (c) The time evolution of the solution (should stay constant)
- (d) The numerical dissipation and introduction of solution oscillations.
- (e) Determine whether or not the scheme is conservative by summing the pollutant concentration over the domain at each time step. Is the total pollutant mass conserved?

An example of the centerline plot that might be generated at different timesteps is shown in figure 3.

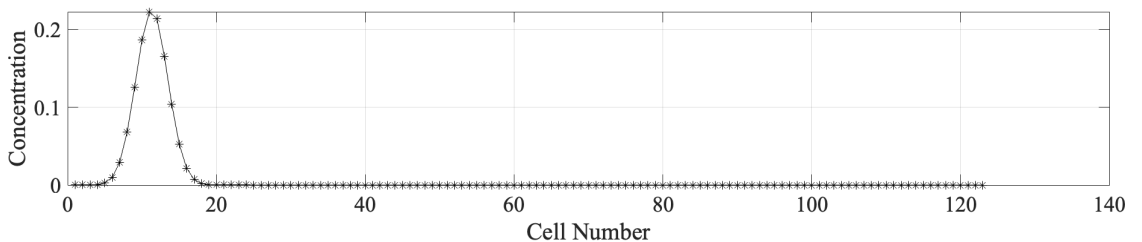


Figure 3: A snapshot of the pollution blob at time $t = 0$ for the midline of the domain ($y = 0.5, x = [0, 6].$)

9. **Removing Pollutant using a nearby sink/pump (10 pts)** Before you can complete this question you will need to update make a copy of your code:

- (a) Create a *new copy* (or plan carefully how to modify your working code) of your working MATLAB FVM code from previous sections.
- (b) Modify the bottom (south-most) and top (north-most) boundaries so that if:
 - The cell-face velocity on that cell boundary is leaving the domain, the pollutant will leave (and not re-enter) the domain.
 - The cell-face velocity on that cell boundary is entering the domain, the flow entering has zero pollutant concentration.
 - Update the velocity field within the domain to reflect the addition of a simple pump model (we will use a potential flow sink located outside the domain at $x = 2, y = 2$). The velocity at any face of the domain can be calculated as the superposition of the groundwater flow (2 units/sec) and the potential sink velocity contribution, which results in the x - and y - velocity expressions below:

$$u(x, y) = 2 - m \frac{x - 2}{(x - 2)^2 + (y - 2)^2} \quad (15)$$

$$v(x, y) = -m \frac{y - 2}{(x - 2)^2 + (y - 2)^2} \quad (16)$$

Determine and plot the total pollutant in the domain vs. time for several different pump flow rates m (suggestion: Use values of m between 0.05 and 1.00).

Question Deliverable: This question is supposed to be a fun application. Show a mini-study of the pumping flow rate vs. total pollutant over time in the domain. Discuss briefly.

The flowfield due to a sink (pump) will look something similar to Figure 4.

3.1 BONUS (10 pts): Implementing and Studying the Diffusive Flux

1. Do a little research and determine a diffusive flux that could be added to the FVM to allow the solution for the convection-diffusion equation.
2. Implement a set of diffusion fluxes.
3. Use a diffusion coefficient of $D = 0.005$.
4. Repeat the study you performed when comparing the upwind and Lax-Wendroff fluxes, but now including the diffusion fluxes.
5. Discuss how your inclusion of diffusion alters the pollutant distribution over time.

3.2 BONUS (10 pts): Implementing the Flux Differencing Form of the Lax-Wendroff Method

If you choose to perform this bonus, you will need to update the fluxes as well as the boundaries to allow for indices to work.

In the Lax-Wendroff approach from earlier, we used a 1-D Lax-Wendroff scheme with simple **Dimensional Splitting** to allow us to implement (with some error) a better 1-D flux function in 2-dimensions. It turns out, there is a 2D Lax-Wendroff flux function which can be implemented, with

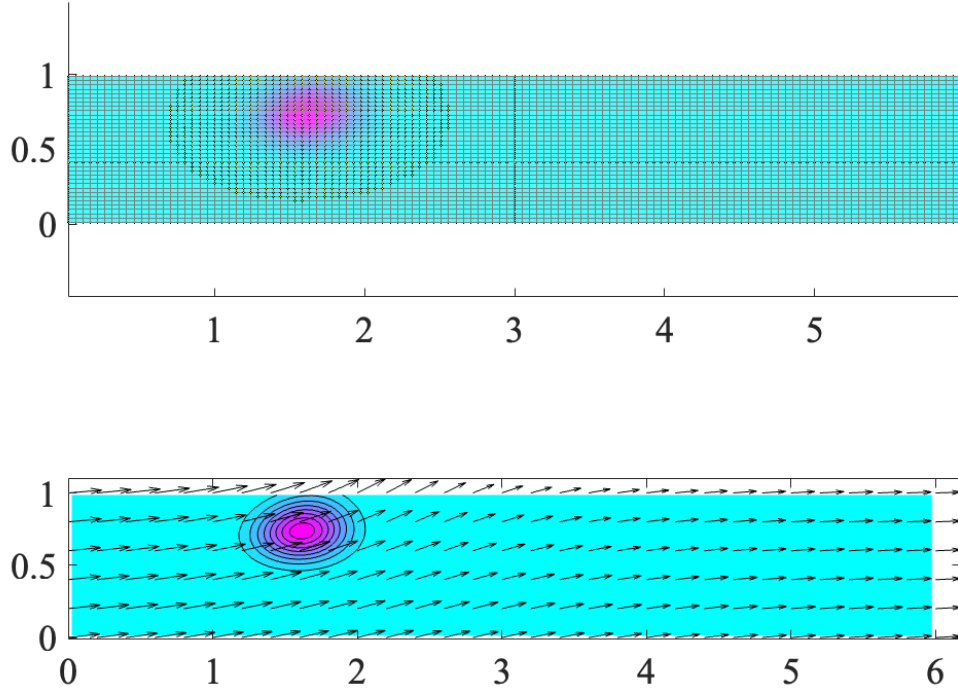


Figure 4: A snapshot of the convected and diffused pollution blob when a moderately strong sink (pump) is included at $x = 2$ and $y = 2$.

cross-difference terms added to better capture the effects of travelling across the grid in combined x - and y motion (see chapter 19, equation 19.14 of Leveque):

$$\begin{aligned}
 F_{i,j-1/2}^y &= V \left(\frac{\phi_{i,j-1} + \phi_{i,j}}{2} \right) - V^2 \frac{\Delta t}{2\Delta y} (\phi_{i,j} - \phi_{i,j-1}) \\
 &\quad - \frac{\Delta t}{8\Delta y} U \cdot V [(\phi_{i+1,j} - \phi_{i,j}) + (\phi_{i+1,j-1} - \phi_{i,j-1}) + (\phi_{i,j} - \phi_{i-1,j}) + (\phi_{i,j-1} - \phi_{i-1,j-1})]
 \end{aligned}$$

$$\begin{aligned}
 F_{i,j-1/2}^y &= V \left(\frac{\phi_{i,j-1} + \phi_{i,j}}{2} \right) - V^2 \frac{\Delta t}{2\Delta y} (\phi_{i,j} - \phi_{i,j-1}) \\
 &\quad - \frac{\Delta t}{8\Delta y} U \cdot V [(\phi_{i+1,j} - \phi_{i,j}) + (\phi_{i+1,j-1} - \phi_{i,j-1}) + (\phi_{i,j} - \phi_{i-1,j}) + (\phi_{i,j-1} - \phi_{i-1,j-1})]
 \end{aligned}$$

If you choose to accept this bonus challenge, please implement this Flux-Differencing form of Lax-Wendroff and perform a study to compare your results to those calculated using the dimensional-splitting version. Note, you should revert to periodic boundary conditions and set a velocity field ($U = 2$ and $V = 1.5$).

3.3 References

Randall Leveque, *Finite Volume Methods for Hyperbolic Problems*, Cambridge Texts in Applied Mathematics, 2002.